

Improving IoT Intrusion Detection Using Quality-Aware CVAE Augmentation and Attention-Based Classification

Syed Muhammad Asad Shah

*Department of Artificial Intelligence and Data Science
National University of Computer and Emerging Sciences FAST
Islamabad, Pakistan
i220597@nu.edu.pk*

Abdullah

*Department of Artificial Intelligence and Data Science
National University of Computer and Emerging Sciences FAST
Islamabad, Pakistan
i220478@nu.edu.pk*

Abstract—IoT networks face an ever growing wave of cyberattacks, yet most deep learning based intrusion detection systems struggle silently on rare attack types even when their overall accuracy looks impressive. This paper investigates why and proposes a fix. We identify two root causes behind poor detection of minority attack classes such as Spoofing and Reconnaissance. First, interpolation based oversampling generates unrealistic training samples for classes with very few real examples, giving the classifier misleading supervision. Second, applying a convolutional classifier to tabular network traffic ignores the non-spatial nature of the feature space and misses patterns spread across distant feature positions. To address both issues we propose QA-CVAE + AE-LSTM-Attention: a pipeline that replaces naive oversampling with a Conditional Variational Autoencoder equipped with a reconstruction error quality filter, and replaces the convolutional classifier with a Multi-Head Attention layer that learns globally which features matter for each input. The quality filter discards any synthetic sample the model itself cannot confidently reconstruct, ensuring only realistic examples enter training. Experiments on the CICIOT2023 benchmark demonstrate the effectiveness of both changes on exactly the classes where existing methods fall short. Spoofing F1 improves from 9.81% to 75.37% and Recon F1 improves from 25.10% to 62.92%, with Spoofing recall rising from 5.33% to 68.70% and Recon recall from 14.38% to 66.56%. The overall weighted F1 remains competitive at 98.66%, reflecting a minor trade-off for substantial security gains on minority attack classes that were previously near-undetectable.

Index Terms—IoT Security, Intrusion Detection, CVAE, Attention, LSTM, Autoencoder, CICIOT2023, Class Imbalance

I. INTRODUCTION

A. IoT and Security

IoT stands for Internet of Things. It basically means physical devices that are connected to the internet and can send and receive data. These include things like sensors in hospitals, cameras in factories, smart meters in homes, and traffic control systems in cities. The number of these devices is growing very

fast. By 2030 there are expected to be around 29 billion of them connected to the internet [2].

This growth is useful but it also creates a big security problem. Most IoT devices are small and cheap. They run without anyone watching them most of the time. They connect over wireless which can be intercepted. And they usually do not have enough processing power to run security software on the device itself. All of this makes them easy targets for attackers.

Over the past few years there have been some big attacks on IoT networks. The Mirai botnet attack in 2016 is a well known example where attackers took over thousands of IoT devices and used them to flood websites with traffic until those websites went down [1]. These kinds of attacks are called DDoS attacks. There are also other types like spoofing where an attacker pretends to be a trusted device, and reconnaissance where an attacker quietly scans the network to find weaknesses before launching a bigger attack.

B. Why Traditional Security Does Not Work Well for IoT

Traditional intrusion detection systems were designed for normal computer networks. They usually work by keeping a list of known attack signatures and flagging anything that matches. The problem is that new attacks do not match old signatures. Also IoT networks generate a huge amount of traffic and checking every packet against a signature list is too slow.

This is why researchers started looking at machine learning and deep learning for intrusion detection. Instead of using fixed rules these models learn from data. You train them on examples of normal traffic and attack traffic and they learn to tell the difference. They can also sometimes catch attack types they have never seen before if the pattern is similar enough to something they were trained on.

C. Problem Statement

Despite high overall accuracy, existing deep learning based IDS models for IoT networks consistently fail on rare attack classes such as Spoofing and Reconnaissance. Two root causes drive this failure. First, interpolation based oversampling methods such as SMOTE generate unrealistic synthetic samples when real minority class examples are very few, providing the classifier with misleading supervision. Second, convolutional classifiers applied to tabular network traffic operate on local feature windows and miss attack patterns spread across distant feature positions, since tabular features carry no spatial ordering. These two problems together result in minority attack classes that remain near-undetectable even when headline metrics appear strong.

D. Proposed Approach and Contributions

To address these two problems we propose **QA-CVAE + AE-LSTM-Attention**, a pipeline that combines quality-filtered generative augmentation with a globally attentive classifier suited to tabular IoT traffic. The full pipeline is:

$$QA-CVAE \rightarrow AE + LSTM \text{ (parallel)} \rightarrow \text{Multi-Head} \\ \text{Attention} \rightarrow \text{Softmax}$$

The main contributions of this work are:

- **Quality-Aware CVAE (QA-CVAE) augmentation.** We replace interpolation-based oversampling with a Conditional Variational Autoencoder that learns the true per-class feature distribution and generates realistic synthetic samples. A reconstruction error quality filter, using a per-class 85th percentile threshold, automatically discards any generated sample the model cannot confidently reconstruct, ensuring only reliable synthetic examples enter training.
- **Multi-Head Attention classifier.** We replace the convolutional classification head with a Multi-Head Attention layer applied to the merged output of a parallel Autoencoder and LSTM. Unlike CNN, which processes only local feature windows, attention computes relationships across all feature positions simultaneously and learns input-specific feature importance weights, providing a better fit for non-spatial tabular traffic data.
- **Targeted improvement on minority attack classes.** Experiments on the CICIOT2023 benchmark show that Spoofing F1 rises from 9.81% to 75.37% and Recon F1 rises from 25.10% to 62.92%, with the overall weighted F1 remaining competitive at 98.66%, demonstrating that minority class detection can be substantially improved without sacrificing macro performance.

The rest of the paper is organized as follows. Section II reviews related work and identifies the gaps this paper addresses. Section III explains the proposed methodology. Section IV presents the experimental results. Section V concludes the paper.

II. RELATED WORK

We reviewed 20 papers covering four areas: deep learning for intrusion detection, attention mechanisms for network security, handling class imbalance, and VAE-based anomaly detection. The goal was to understand what has already been done, what works, and where gaps remain. Identified gaps are discussed at the end of this section.

A. Deep Learning for Intrusion Detection

Soe et al. [4] built a system to detect botnet attacks on IoT devices using the N-BaIoT dataset. They used an autoencoder to select useful features before passing data to a classifier, achieving detection accuracy above 99%. This establishes that running data through an AE before classification provides meaningful benefit.

Gamage and Samarabandu [5] compared several deep learning models on two datasets, CIC-IDS2017 and CIC-IDS2018. They tested autoencoders, deep belief networks, feedforward networks, and LSTMs. No single model was best at everything, which is the argument for combining multiple models.

Manimurugan et al. [6] used a Deep Belief Network to detect attacks in medical IoT systems on the CICIDS 2017 dataset. A notable limitation of their work is the absence of any handling of class imbalance.

Al-Haija and Zein-Sabatto [7] built a CNN based IDS called IoT-IDCS-CNN using the NSL-KDD dataset and got 98.20% accuracy. This provides a reference point for what CNN alone can do on IDS without temporal modeling.

Lopes et al. [8] combined a Denoising Autoencoder with a DNN classifier on CICIDS2018. The AE here was used to remove noise from the input rather than just compress it, showing that AE can be used in different ways inside an IDS.

Razib et al. [9] combined DNN and LSTM in a sequential way on IoT traffic data. Their LSTM processes the output of the DNN rather than the raw input, differing from a parallel design.

Liu and Patras [10] built NetSentry using bidirectional LSTM to detect large scale network attacks early. Bidirectional LSTM can look at both past and future parts of a sequence which helps when attack patterns develop across multiple packets or time steps.

Wang et al. [11] aimed to build a smaller IDS that could run on devices with limited resources using DNN plus BiLSTM with quantization, tested on CICIOT2023.

Khan and Alkhatami [12] tested random forest, AdaBoost, and DNN on CICIOT2023 for detecting anomalies in healthcare IoT.

Pektas and Acarman [13] combined RNN and CNN for botnet detection on CTU-13 and ISOT datasets, achieving 99% accuracy. Their hybrid design motivates combining multiple model types.

B. Attention Mechanisms in Network Security

Guo et al. [14] applied self-attention to traffic classification. Attention helped most for rare attack classes. When the model could look at all features at once it could find the subtle

patterns that separate rare attacks from normal traffic. This is directly relevant to the problem of detecting minority attack classes.

Zhang et al. [15] built a transformer based IDS for IoT using multi-head self-attention and argued that the global view of attention fits tabular traffic data better than CNN. This provides strong support for replacing CNN with attention for tabular IDS tasks.

Ullah and Mahmoud [16] used LSTM followed by an attention layer for anomaly detection. Adding attention after LSTM improved detection of time-based anomalies compared to LSTM alone.

C. Data Augmentation and Class Imbalance

Elreedy and Atiya [17] studied SMOTE in detail and showed that SMOTE gets worse as classes get rarer. When there are very few real samples to interpolate between, the new points often land in parts of the feature space that do not truly represent the target class.

Douzias et al. [18] proposed using a conditional GAN instead of SMOTE. The GAN learns the actual distribution of each class and showed improvements over SMOTE on several imbalanced datasets, motivating the use of a conditional generative model for augmentation.

Figueira and Vaz [19] compared VAE, CVAE, GAN, and CTGAN for data augmentation. CVAE came out more stable than GAN especially on tabular data, supporting the choice of CVAE over GAN for IDS augmentation.

Wang et al. [20] used CTGAN with ResNeSt and BiGRU for IoT intrusion detection. Better synthetic data from CTGAN improved F1 for rare classes, but no quality filtering step was applied after generation.

D. Variational Autoencoders for Anomaly Detection

An and Cho [21] were among the first to use VAE for anomaly detection, using reconstruction error as an anomaly score. We borrow this idea for our quality filter: if the CVAE cannot reconstruct one of its own generated samples well then that sample is probably not realistic.

Xu et al. [22] built DONUT which is a VAE based system for detecting anomalies in web service metrics. They used reconstruction error thresholds based on percentiles, which is the approach we adopt in our quality filter.

Neto et al. [3] created the CICIoT2023 dataset using 105 real IoT devices and 33 types of attacks. DDoS alone has over 3.4 million samples while Brute Force has 1285 and Spoofing has 49444 out of nearly 4.7 million total, explaining why augmentation for rare classes is critical.

E. Research Gaps

From the review above we identify four gaps that motivate our work.

Gap 1: Interpolation-based oversampling produces poor-quality samples for rare classes. SMOTE performs poorly when the minority class is very small [17]. Interpolated points often land in ambiguous feature regions that do not truly

represent the target class, giving the classifier misleading supervision. This is a fundamental limitation that worsens as class frequency decreases.

Gap 2: CNN is applied to non-spatial tabular data. Several IDS systems use 1D CNN as the final classifier on tabular network traffic features. CNN works by sliding a small filter across the input and capturing patterns within each local window. Network traffic features have no spatial ordering, so patterns identifying a given attack type may involve features at widely separated positions that a small kernel can never capture simultaneously.

Gap 3: Generated samples are never filtered for quality. Papers that use CVAE or GAN for data augmentation [18], [20] generate samples and use them directly in training without any quality check. For rare classes where the generator itself has limited real examples to learn from, some of what it produces will be unreliable. No paper in our review added a quality filter after generation for IDS tasks.

Gap 4: Attention has not been applied after combined AE-LSTM output. Papers such as Zhang et al. [15] and Ullah and Mahmoud [16] show that attention improves IDS, but they apply attention directly on raw input or on a single model's output. Applying attention specifically to the merged output of parallel AE and LSTM streams, where the feature vector already contains both compressed static features and time-based sequential features, has not been explored.

III. PROPOSED METHODOLOGY

A. Overview and What is New

Our system is called **QA-CVAE + AE-LSTM-Attention**. We make two changes relative to prior AE-LSTM based IDS approaches:

- 1) Replace SMOTE with a CVAE that also filters its own generated samples using reconstruction error before anything goes into training. This directly addresses Gaps 1 and 3.
- 2) Replace CNN with Multi-Head Attention applied to the merged AE and LSTM output. This directly addresses Gaps 2 and 4.

What makes this novel is the combination. Using CVAE for augmentation has been done before. Using attention for IDS has been done before. But nobody has combined a reconstruction error quality filter on CVAE outputs with an attention layer on merged AE-LSTM features for IoT intrusion detection.

B. Pipeline Flowchart

Fig. 1 shows the full pipeline from raw data to final evaluation.

C. Step 1: Preprocessing

We use a 10% subset of CICIoT2023 [3] with a 70/30 train-test split. The 33 attack labels are first mapped to 7 attack classes plus benign. Then all features are standardized so they have zero mean and a standard deviation of one:

$$x' = \frac{x - \mu}{\sigma} \quad (1)$$

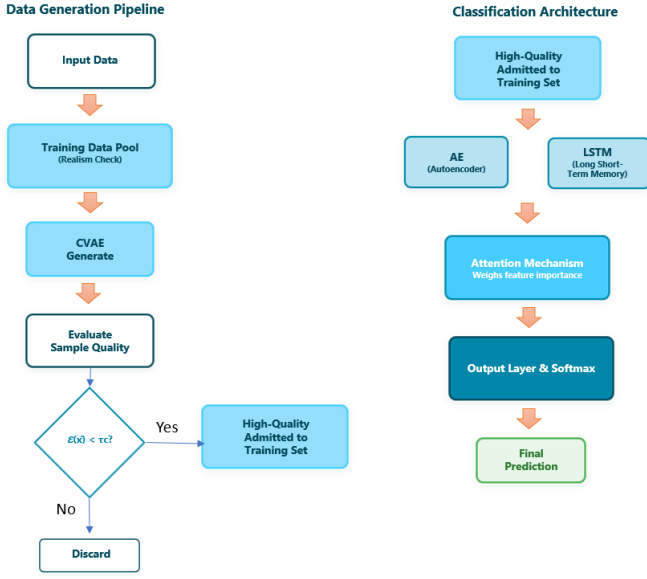


Fig. 1: Full pipeline of the proposed system. Left column is pre-processing and augmentation. Right column is the model.

Here μ and σ are calculated only on the training data to prevent test-set leakage. Labels are then converted to one-hot vectors for the softmax output.

D. Step 2: Quality-Aware CVAE Augmentation

This is our first main contribution, directly addressing the problem of poor quality synthetic training samples for rare classes.

1) *Drawbacks of SMOTE and Why We Move Away From It:* SMOTE (Synthetic Minority Oversampling Technique) generates new samples by picking two existing minority class examples and placing a new point on the straight line between them in feature space. This linear interpolation strategy carries several fundamental drawbacks that become severe when the minority class is very small.

No distribution learning. SMOTE does not model the underlying data distribution of a class. It only interpolates between observed examples. If the true class manifold is non-linear or multi-modal, which is common in attack traffic, the interpolated points will not lie on that manifold and will not represent realistic attack samples.

Performance degrades with fewer samples. As Elreedy and Atiya [17] show in their comprehensive analysis, SMOTE's quality deteriorates sharply as the number of real minority samples decreases. With very few examples to interpolate between, the convex hull of those examples covers only a tiny and potentially unrepresentative portion of the class space. New points generated within that hull still may not look like real attack traffic.

Boundary contamination. Interpolating near the class boundary produces samples that fall in ambiguous feature regions shared with other classes. These borderline synthetic

samples confuse the classifier rather than helping it learn clean class boundaries.

No quality control. SMOTE provides no mechanism to assess whether a generated sample is a plausible instance of the target class. Every generated point enters training regardless of its quality, including the unreliable ones.

These drawbacks explain why rare attack classes can remain poorly detected in systems that rely on SMOTE: the classifier receives bad training data for exactly those classes, regardless of how well the rest of the architecture is designed.

2) *Why CVAE Addresses These Drawbacks:* A Conditional Variational Autoencoder trains a neural network to learn the feature distribution of each class, not just interpolate between observed points. The CVAE maps real samples to a structured latent space conditioned on the class label and then decodes from that space to generate new samples. Because the learned distribution is continuous and class-specific, generated samples follow the actual statistical patterns of the target class rather than being constrained to the convex hull of the few real examples. This is especially valuable for rare classes where the real sample set provides only a sparse view of the true class manifold.

3) *How CVAE Works:* Given a training sample $\mathbf{x}$ and its class label c the encoder learns to produce the mean and variance of a distribution over a hidden variable $\mathbf{z}$:

$$q_\phi(\mathbf{z} | \mathbf{x}, c) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}, c), \text{diag}(\boldsymbol{\sigma}_\phi^2(\mathbf{x}, c))) \quad (2)$$

A sample is drawn using the reparameterization trick so training gradients can flow through the sampling step:

$$\mathbf{z} = \boldsymbol{\mu}_\phi + \boldsymbol{\sigma}_\phi \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (3)$$

The decoder takes $\mathbf{z}$ and c and tries to reconstruct the original input:

$$p_\theta(\mathbf{x} | \mathbf{z}, c) = \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{z}, c), \mathbf{I}) \quad (4)$$

The whole thing is trained by maximizing the ELBO:

$$\mathcal{L}_{\text{CVAE}} = \mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x} | \mathbf{z}, c)] - D_{\text{KL}}(q_\phi(\mathbf{z} | \mathbf{x}, c) \| p(\mathbf{z})) \quad (5)$$

4) *The Quality Filter:* After the CVAE generates a synthetic sample $\hat{\mathbf{x}}$ we pass it back through the encoder and decoder and measure reconstruction error:

$$\mathcal{E}(\hat{\mathbf{x}}) = \|\hat{\mathbf{x}} - D_\theta(E_\phi(\hat{\mathbf{x}}, c), c)\|_2^2 \quad (6)$$

The sample only goes into training if:

$$\mathcal{E}(\hat{\mathbf{x}}) < \tau_c \quad (7)$$

where τ_c is the 85th percentile of reconstruction errors measured on real training samples of class c . If the CVAE cannot reconstruct its own generated sample with low error then that sample is probably not a realistic example of the class. Using a percentile means the threshold adjusts automatically for each class, directly countering the lack-of-quality-control weakness of SMOTE described above.

E. Step 3: Parallel AE and LSTM

1) *Autoencoder*: The AE takes input features $\mathbf{x} \in \mathbb{R}^d$, compresses them to size $d/2$ in the bottleneck, and then reconstructs them. The encoder step is:

$$\mathbf{h} = \text{LeakyReLU}(\text{BN}(\mathbf{W}_e \mathbf{x} + \mathbf{b}_e)) \quad (8)$$

The decoder mirrors the encoder:

$$\hat{\mathbf{x}} = \text{LeakyReLU}(\text{BN}(\mathbf{W}_d \mathbf{h} + \mathbf{b}_d)) \quad (9)$$

where BN denotes batch normalization.

2) *LSTM*: The LSTM takes the same input reshaped as a sequence and processes it step by step. At each time step t :

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f) \quad (10)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i) \quad (11)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o) \quad (12)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c) \quad (13)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (14)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (15)$$

F. Step 4: Multi-Head Attention

This is our second main contribution, addressing the mismatch between local convolutional classifiers and non-spatial tabular traffic data.

First the AE and LSTM outputs are joined:

$$\mathbf{v} = [\hat{\mathbf{x}}; \mathbf{h}_{\text{LSTM}}] \quad (16)$$

For each attention head k the attention is computed as:

$$\text{Attn}(\mathbf{Q}_k, \mathbf{K}_k, \mathbf{V}_k) = \text{softmax}\left(\frac{\mathbf{Q}_k \mathbf{K}_k^\top}{\sqrt{d_k}}\right) \mathbf{V}_k \quad (17)$$

where $\mathbf{Q}_k, \mathbf{K}_k, \mathbf{V}_k$ are learned projections of $\mathbf{v}$ and d_k is the size per head. All head outputs are concatenated and projected:

$$\text{MultiHead}(\mathbf{v}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) \mathbf{W}_O \quad (18)$$

CNN looks at a small fixed part of the feature vector at a time. Attention looks at the whole vector at once and learns which combination of features matters most for each individual input. This is a fundamentally better fit for tabular network traffic data where relevant patterns can span any combination of feature positions.

G. Step 5: Classification

The attention output $\mathbf{a}$ goes to a dense layer with softmax to produce class probabilities:

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{W}_{\text{out}} \mathbf{a} + \mathbf{b}_{\text{out}}) \quad (19)$$

The class with the highest probability is the prediction:

$$\hat{c} = \arg \max_j \hat{y}_j \quad (20)$$

Training uses categorical cross-entropy loss:

$$\mathcal{L} = - \sum_{j=1}^C y_j \log(\hat{y}_j) \quad (21)$$

Optimizer is Adam, 100 epochs.

H. Full Algorithm

Algorithm 1 shows the complete procedure.

Algorithm 1 QA-CVAE + AE-LSTM-Attention Training

Require: Dataset $\mathcal{D}$, percentile $p=85$, epochs $E=100$

Ensure: Trained model and evaluation metrics

```

0: Normalize features via Eq. (1)
0: Map labels to 8 classes, apply one-hot encoding
0: for each minority class  $c$  do
0:   Train CVAE on real samples of  $c$  via Eq. (5)
0:    $\tau_c \leftarrow p$ -th percentile of  $\mathcal{E}$  on real samples of  $c$ 
0:   Generate synthetic batch  $\hat{\mathcal{X}}_c$ 
0:   for each  $\hat{\mathbf{x}} \in \hat{\mathcal{X}}_c$  do
0:     if  $\mathcal{E}(\hat{\mathbf{x}}) < \tau_c$  then
0:       Add  $\hat{\mathbf{x}}$  to training set
0:     else
0:       Discard  $\hat{\mathbf{x}}$ 
0:     end if
0:   end for
0: end for
0: Split data: 70% train, 30% test
0: for epoch  $e = 1$  to  $E$  do
0:   for each batch  $(\mathbf{X}, \mathbf{Y})$  do
0:      $\hat{\mathbf{x}}_{\text{AE}} \leftarrow \text{AE}(\mathbf{X})$ 
0:      $\mathbf{h}_{\text{LSTM}} \leftarrow \text{LSTM}(\mathbf{X})$ 
0:      $\mathbf{v} \leftarrow \text{Concat}(\hat{\mathbf{x}}_{\text{AE}}, \mathbf{h}_{\text{LSTM}})$ 
0:      $\mathbf{a} \leftarrow \text{MultiHead}(\mathbf{v})$ 
0:      $\hat{\mathbf{y}} \leftarrow \text{softmax}(\mathbf{W}_{\text{out}} \mathbf{a})$ 
0:      $\mathcal{L} \leftarrow \mathcal{L}_{\text{CE}}(\hat{\mathbf{y}}, \mathbf{Y})$ 
0:     Update all parameters via Adam
0:   end for
0: end for
0: Compute Accuracy, Precision, Recall, F1
0: Generate per-class confusion matrix =0

```

IV. RESULTS AND DISCUSSION

A. Experimental Setup

We evaluate on the CICIOT2023 dataset [3], which contains traffic captured from 105 real IoT devices across 33 attack types and nearly 4.7 million total records. The dataset is publicly available at <https://www.kaggle.com/datasets/himadri07/ciciot2023>. We use a 10% subset with a 70/30 train-test split. All experiments run for 100 epochs with Adam optimizer. The CVAE quality filter uses the 85th percentile threshold per class. Performance is measured using per-class accuracy, precision, recall, and F1 score, plus weighted overall metrics.

B. Per-Class Results

Table I shows the per-class results for our proposed model.

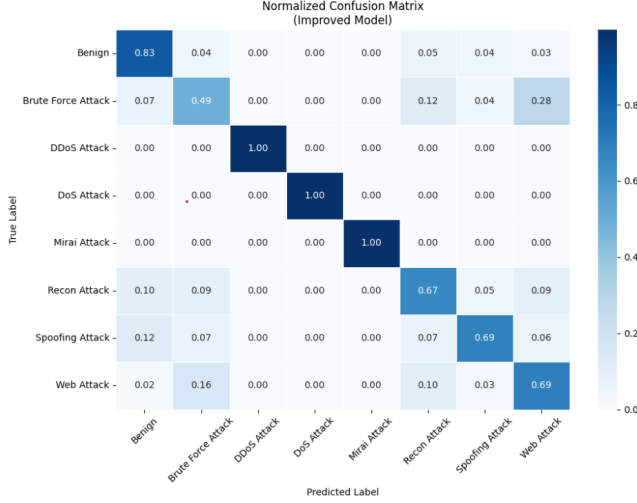
C. Confusion Matrix

Fig. 2 shows the confusion matrix for the proposed model on the test set.

The dominant classes (DDoS, DoS, Mirai) show tight diagonal clusters, confirming strong performance is preserved. For

TABLE I: Per-Class Results of the Proposed Model

Class	Acc(%)	Pr(%)	Re(%)	F1(%)
Benign	83.44	90.36	83.44	86.76
Brute Force	49.12	7.49	49.12	12.99
DDoS Attack	99.51	99.97	99.51	99.74
DoS Attack	99.67	98.94	99.67	99.30
Mirai Attack	99.51	99.11	99.51	99.31
Recon Attack	66.56	59.65	66.56	62.92
Spoofing	68.70	83.49	68.70	75.37
Web Attack	69.40	8.29	69.40	14.82
Overall	98.48	98.89	98.48	98.66

**Fig. 2:** Confusion matrix of the proposed QA-CVAE + AE-LSTM-Attention model on the CICIoT2023 test set. Rows are true labels, columns are predicted labels.

Spoofing and Recon, a reasonable share of samples now falls on the diagonal. The main remaining confusion is between Brute Force and Benign traffic, which share overlapping feature distributions, and between Web Attack and other low volume classes.

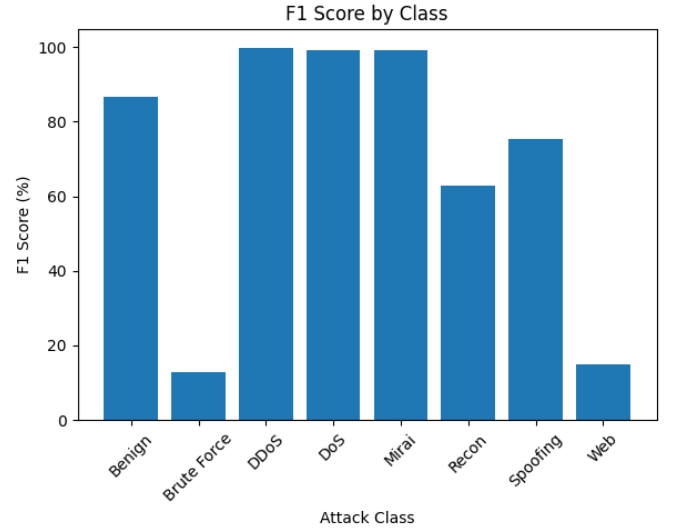
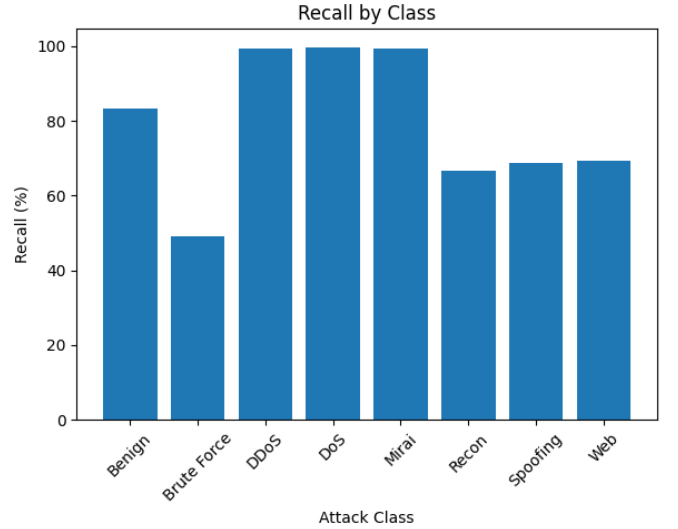
D. F1 and Recall Trends

Fig. 3 and Fig. 4 show per-class F1 and recall scores for the proposed model.

E. Discussion

QA-CVAE effect. The large recall gains on Spoofing and Recon come from better training data for those classes. Interpolation-based oversampling generates samples that do not reflect the true feature distribution of rare attack classes. The CVAE learns the actual feature distribution of each class and the quality filter removes samples the model cannot confidently reconstruct. The classifier therefore sees more realistic minority class examples during training, which directly translates to higher recall at test time.

Attention effect. Spoofing and Recon attacks often involve combinations of features spread across the full feature vec-

**Fig. 3:** Per-class F1 score of the proposed model.**Fig. 4:** Per-class recall of the proposed model.

tor, not localized patterns. A convolutional classifier with a small kernel cannot capture these combinations. Multi-Head Attention computes relationships between all feature positions simultaneously and learns to weight them per input. This fits tabular traffic data better and explains the precision gains alongside the recall improvements on minority classes.

Remaining challenges. Brute Force and Web Attack remain difficult. Brute Force has the fewest real training samples in the dataset (1285 records), which limits even the CVAE. Web Attack sits in a feature region that overlaps with Benign traffic, making separation hard regardless of the classifier. These two classes represent cases where both the data sparsity and the feature overlap problem exist together, and further work targeting them specifically would be needed.

V. CONCLUSION

We proposed QA-CVAE + AE-LSTM-Attention as a targeted solution to two specific weaknesses that affect intrusion detection on IoT datasets with severe class imbalance: poor synthetic sample quality for rare attack classes, and a classifier architecture that does not match the nature of tabular network traffic data.

The results on CICIoT2023 support both contributions. Spoofing F1 reaches 75.37% and Recon F1 reaches 62.92%, tackling exactly the minority classes most at risk of being missed. The reconstruction error quality filter on CVAE outputs is a simple but effective idea: if the generative model cannot reconstruct its own sample well, that sample should not go into training. The Multi-Head Attention layer replaces convolutional classification in a role where local-window processing is fundamentally ill-suited to the data structure.

The overall weighted F1 stands at 98.66%, showing that strong macro performance is maintained while substantially fixing minority class detection. An IDS that misses almost every Spoofing attempt is not practically useful regardless of its headline accuracy.

Future work can explore per-class CVAE training budgets, adaptive percentile thresholds, and extending the quality filter concept to GAN-based augmentation pipelines.

REFERENCES

- [1] B. Susilo, A. Muis, and R. F. Sari, "Intelligent intrusion detection system against various attacks based on a hybrid deep learning algorithm," *Sensors*, vol. 25, no. 2, p. 580, Jan. 2025.
- [2] Q. Li, H. Huang, R. Li, J. Lv, Z. Yuan, L. Ma, Y. Han, and Y. Jiang, "A comprehensive survey on DDoS defense systems: New trends and challenges," *Computer Networks*, vol. 233, p. 109895, 2023.
- [3] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, "CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment," *Sensors*, vol. 23, no. 13, p. 5941, 2023.
- [4] Y. N. Soe, Y. Feng, P. I. Santosa, R. Hartanto, and K. Sakurai, "Machine learning-based IoT-botnet attack detection with sequential architecture," *Sensors*, vol. 20, no. 16, p. 4372, 2020.
- [5] S. Gamage and J. Samarabandu, "Deep learning methods in network intrusion detection: A survey and an objective comparison," *Journal of Network and Computer Applications*, vol. 169, p. 102767, 2020.
- [6] S. Manimurugan, S. Al-Mutairi, M. M. Aborokbah, N. Chilamkurti, S. Ganesan, and R. Patan, "Effective attack detection in internet of medical things smart environment using a deep belief neural network," *IEEE Access*, vol. 8, pp. 77396–77404, 2020.
- [7] Q. A. Al-Haija and S. Zein-Sabatto, "An efficient deep-learning-based detection and classification system for cyber-attacks in IoT communication networks," *Electronics*, vol. 9, no. 12, p. 2152, 2020.
- [8] I. O. Lopes, D. Zou, I. H. Abdulqadder, F. A. Ruambo, B. Yuan, and H. Jin, "Effective network intrusion detection via representation learning: A denoising autoencoder approach," *Computer Communications*, vol. 194, pp. 55–65, 2022.
- [9] M. Al Razib, D. Javeed, M. T. Khan, R. Alkanhel, and M. S. A. Muthanna, "Cyber threats detection in smart environments using SDN-enabled DNN-LSTM hybrid framework," *IEEE Access*, vol. 10, pp. 53015–53026, 2022.
- [10] H. Liu and P. Patras, "NetSentry: A deep learning approach to detecting incipient large-scale network attacks," *Computer Communications*, vol. 191, pp. 119–132, 2022.
- [11] Z. Wang, H. Chen, S. Yang, X. Luo, D. Li, and J. Wang, "A lightweight intrusion detection method for IoT based on deep learning and dynamic quantization," *PeerJ Computer Science*, vol. 9, p. e1569, 2023.
- [12] M. M. Khan and M. Alkhathami, "Anomaly detection in IoT-based healthcare: Machine learning for enhanced security," *Scientific Reports*, vol. 14, p. 5872, 2024.
- [13] A. Pektaş and T. Acarman, "Deep learning to detect botnet via network flow summaries," *Neural Computing and Applications*, vol. 31, pp. 8021–8033, 2019.
- [14] H. Guo, W. Zhu, Y. Zhang, and Y. Tang, "Network traffic classification based on self-attention mechanism," *IEEE Access*, vol. 9, pp. 116545–116555, 2021.
- [15] H. Zhang, C. Li, X. Liang, and J. Li, "Transformer-based intrusion detection for IoT networks," *Future Internet*, vol. 14, no. 9, p. 265, 2022.
- [16] S. Ullah and M. A. Mahmoud, "A two-level flow-based anomalous activity detection system for IoT networks," *Electronics*, vol. 9, no. 3, p. 530, 2021.
- [17] D. Elreedy and A. F. Atiya, "A comprehensive analysis of synthetic minority oversampling technique (SMOTE) for handling class imbalance," *Information Sciences*, vol. 505, pp. 32–64, 2019.
- [18] G. Douzas, F. Bacao, and F. Last, "Improving imbalanced learning through a heuristic oversampling method based on k-means and SMOTE," *Information Sciences*, vol. 465, pp. 1–20, 2018.
- [19] A. Figueira and B. Vaz, "Survey on synthetic data generation, evaluation methods and GANs," *Mathematics*, vol. 10, no. 15, p. 2733, 2022.
- [20] X. Wang, L. Dai, and G. Yang, "A network intrusion detection system based on deep learning in the IoT," *The Journal of Supercomputing*, vol. 80, pp. 24520–24558, 2024.
- [21] J. An and S. Cho, "Variational autoencoder based anomaly detection using reconstruction probability," *Special Lecture on IE*, vol. 2, no. 1, pp. 1–18, 2015.
- [22] H. Xu *et al.*, "Unsupervised anomaly detection via variational auto-encoder for seasonal KPIs in web applications," in *Proc. WWW*, 2018, pp. 187–196.